

Dynamic Linear Models and the Kalman Filter

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The Problem

The Model

The Kalman Filter

The Prior

Implementation

Tuning

Summary



Noisy measurements of a moving target

We want to track something that changes over time, but our measurements have error.

Examples:

- True unemployment rate (surveys have sampling error)
- Patient's blood pressure (instruments have noise)
- Demand for a product (observed sales are noisy)

Goal: Estimate the true underlying value from noisy data.

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The local level model

Let θ_t = true value, Y_t = observation

Observation equation:

$$Y_t = \theta_t + v_t, \quad v_t \sim N(0, V)$$

We see the truth plus measurement noise.

State equation:

$$\theta_t = \theta_{t-1} + w_t, \quad w_t \sim N(0, W)$$

The truth evolves as a random walk.



The two variances

- V = observation noise variance
 - Larger $V \rightarrow$ noisier measurements
- W = state evolution variance
 - Larger $W \rightarrow$ truth can change more

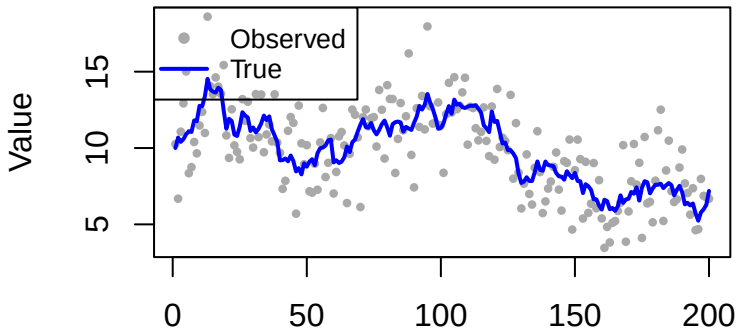
The ratio W/V will determine how the filter behaves.



Simulated example

```
set.seed(42)
n <- 200; V <- 4; W <- 0.25

theta <- cumsum(c(10, rnorm(n-1, 0, sqrt(W)))) # true
Y <- theta + rnorm(n, 0, sqrt(V))               # observed
```



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The idea

At each time t , combine two pieces of information:

1. **Prediction:** Where we think θ_t is (before seeing Y_t)
2. **Observation:** The new data point Y_t

The filter computes a **weighted average**:

$$\text{Estimate} = (1 - K) \times \text{Prediction} + K \times \text{Observation}$$

where K is the **Kalman gain** (between 0 and 1).



The algorithm

Step 1: PREDICT

$$a_t = m_{t-1} \quad \text{(predicted mean)}$$

$$R_t = C_{t-1} + W \quad \text{(predicted variance)}$$

Step 2: UPDATE

$$K_t = \frac{R_t}{R_t + V} \quad \text{(Kalman gain)}$$

$$m_t = a_t + K_t(Y_t - a_t) \quad \text{(updated mean)}$$

$$C_t = (1 - K_t)R_t \quad \text{(updated variance)}$$



Where do the prediction formulas come from?

Recall the state equation: $\theta_t = \theta_{t-1} + w_t$

Predicted mean:

- Our best guess for θ_{t-1} was m_{t-1}
- The expected change is $E[w_t] = 0$
- So our prediction is: $a_t = m_{t-1} + 0 = m_{t-1}$

Predicted variance:

- We were uncertain about θ_{t-1} by C_{t-1}
- The random walk adds uncertainty W
- Variances add: $R_t = C_{t-1} + W$



Where do the update formulas come from?

We have two pieces of information about θ_t :

- Prediction a_t with variance R_t
- Observation Y_t with variance V

Key idea: When combining estimates, weight by precision (1/variance).

Precision of prediction: $1/R_t$

Precision of observation: $1/V$

Total precision: $1/R_t + 1/V$



Deriving the Kalman gain

Weight on observation = (its precision) / (total precision):

$$K = \frac{1/V}{1/R_t + 1/V}$$

Multiply top and bottom by $R_t \cdot V$:

$$K = \frac{R_t}{R_t + V}$$

$$\text{Weight on prediction} = 1 - K = \frac{V}{R_t + V}$$



Deriving the updated mean and variance

Updated mean (weighted average):

$$m_t = (1 - K) \cdot a_t + K \cdot Y_t = a_t + K(Y_t - a_t)$$

Updated variance (combined precision):

$$\frac{1}{C_t} = \frac{1}{R_t} + \frac{1}{V}$$

Solving for C_t :

$$C_t = \frac{R_t \cdot V}{R_t + V} = R_t \cdot \frac{V}{R_t + V} = R_t(1 - K) = (1 - K)R_t$$



The Kalman gain

$$K_t = \frac{R_t}{R_t + V} = \frac{\text{prediction uncertainty}}{\text{total uncertainty}}$$

- $K \approx 1$: Trust the observation (prediction is uncertain)
- $K \approx 0$: Trust the prediction (observation is noisy)

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Starting the filter

The filter needs a **starting point**:

- m_0 = initial guess for the true value
- C_0 = how uncertain we are about that guess

This is called the **prior**—what we believe before seeing any data.



Choosing the prior

If you have prior knowledge:

- Set m_0 to your best guess
- Set C_0 to reflect your confidence

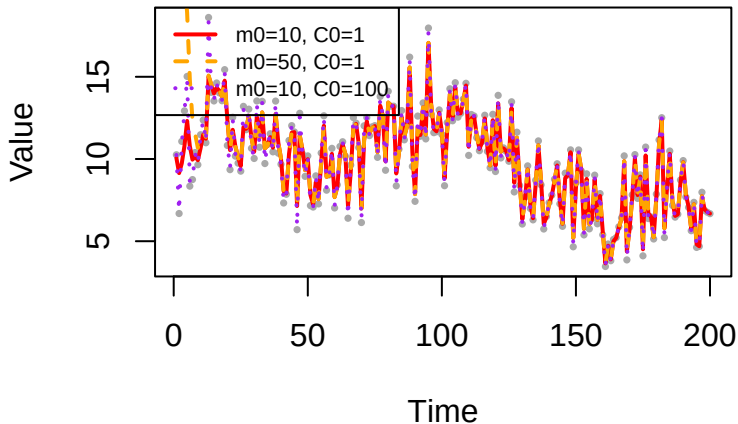
If you don't know:

- Set m_0 to first observation: $m_0 = Y_1$
- Set C_0 large (“diffuse prior”)—says “I’m not sure”

Good news: With enough data, the prior washes out.



Prior doesn't matter much (with enough data)



Different starting points converge to the same estimates.

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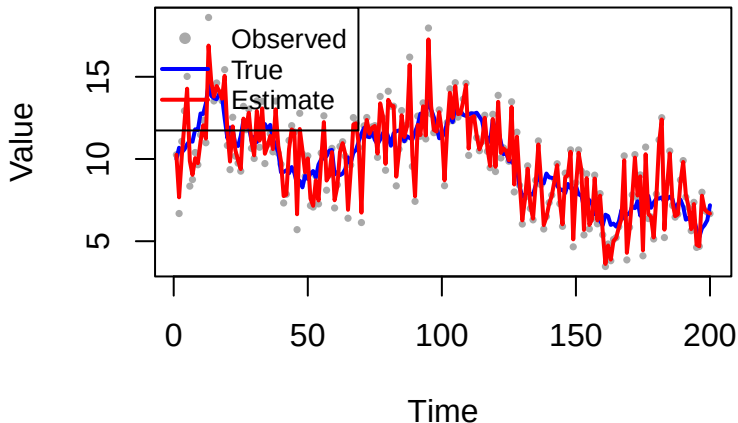
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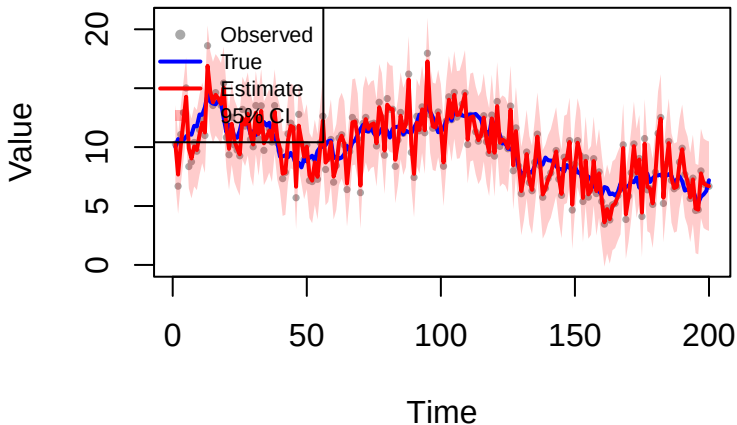
Summary

```
kalman_filter <- function(Y, V, W, m0, C0) {  
  n <- length(Y)  
  m <- C <- numeric(n)  # storage for estimates  
  a <- m0; R <- C0      # initialize  
  
  for (t in 1:n) {  
    # PREDICT  
    R <- R + W  
  
    # UPDATE  
    K <- R / (R + V)  
    m[t] <- a + K * (Y[t] - a)  
    C[t] <- (1 - K) * R  
  
    a <- m[t]  # for next iteration  
  }  
  return(list(m = m, C = C))  
}
```



The filter recovers the true signal from noisy observations.

With uncertainty



The filter also tells us how uncertain we are.

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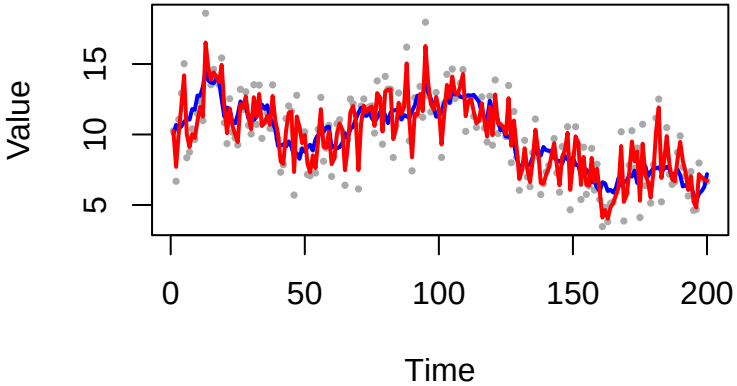
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Small W/V : Smooth estimates

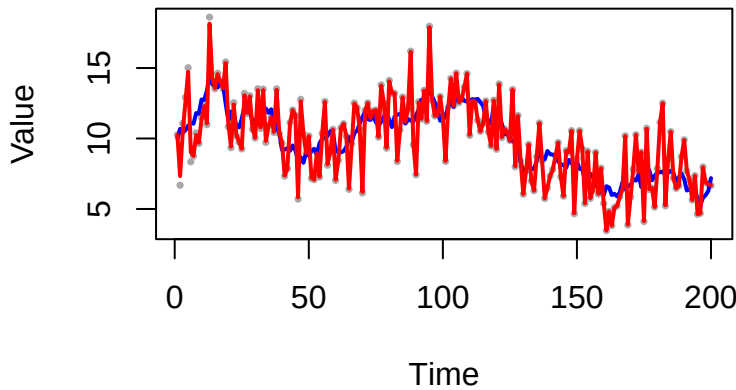
$W/V = 0.0025$ (heavy smoothing)





Large W/V : Responsive estimates

$W/V = 1$ (tracks closely)



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Key points

1. **Model:** Truth is a random walk, observed with noise
2. **Algorithm:** PREDICT $\rightarrow$ UPDATE at each time step
3. **Kalman gain:** Weights prediction vs. observation
4. **Prior:** Starting guess (m_0, C_0) ; washes out with data
5. **Tuning:** W/V ratio controls smoothness



The Kalman filter in one sentence

Optimally combine what you predicted
with what you observed,
weighting by their uncertainties.